Amine Bahi

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Education

École normale supérieure | PhD in Cryptography 2024-2027
Paris, France

- Topics: ORAM and Structured Encryption
- Advisors: Brice Minaud and Tarik Moataz
- Team: Cascade Cryptography Group

Mohammadia School of Engineers 2016-2019
MS in Computer Science
Rabat, Morocco

CPGE | Classes Préparatoires aux Grandes Écoles 2014-2016
Rabat, Morocco
Preparatory Classes

Research Experience

Northeastern University | Research Intern 2023-2024
Boston, USA

- Designed and implemented a low-latency encrypted multi-map (EMM) in C++ with low leakage and subliminal modes; integrated Path ORAM and queueing-theoretic techniques.

UM6P College of Computing | Doctoral Researcher 2020-2023
Marrakesh, Morocco

- Researched searchable symmetric encryption (SSE), structured encryption, and multi-server volume-hiding EMM constructions; evaluated efficiency/leakage trade-offs empirically.
- Implemented and benchmarked the volume-hiding SSE schemes in the research evaluation suite by extending the Clusion encrypted-search framework.
- Studied SSE workload/leakage statistics, Zipf-like keyword distributions, and links to equity-market microstructure; investigated prophet inequalities and secretary-problem optimal stopping.

Inria | Research Intern 03/2019-06/2019
Lyon, France

- Integrated TFHE into an IoT healthcare activity-recognition system; implemented encrypted-inference circuits and optimized them through benchmarking.

Selected Publications

On the Costs of Multi-Server Volume-Hiding
Amine Bahi, Marilyn George, Seny Kamara, Tarik Moataz.
SCN 2026.

Resizable Oblivious RAM
Amine Bahi, Brice Minaud, Tarik Moataz.
EUROCRYPT 2026.

Tigro: Trust Infrastructure for Grassroots Organizing via Grounded Digital Annotations
Leah Namisa Rosenbloom, Seny Kamara, Zachary Espiritu, Tarik Moataz, Amine Bahi, John Wilkinson.
PETS 2026.

Subliminal Encrypted Multi-Maps and Black-Box Leakage Absorption
Amine Bahi, Seny Kamara, Tarik Moataz, Guevara Noubir.
IACR ePrint, 2024.

Privacy-preserving IoT Framework for Activity Recognition in Personal Healthcare Monitoring
Théo Jourdan, Antoine Boutet, Amine Bahi, Carole Frindel.
ACM Trans. Comput. Healthcare, 2021.

Talks and Posters

Resizable Oblivious RAM 13 May 2026
EUROCRYPT, Rome, Italy. Conference presentation.

On the Costs of Multi-Server Volume-Hiding 15 September 2026
SCN, Amalfi, Italy. Conference presentation.

Structured encryption 22 June 2022
UM6P Doctoral Day 2022, Benguerir, Morocco. Presentation and poster.

Resizable Oblivious RAM 17 July 2026
Invited Speaker, MongoDB Crypto Research Talk.

Applied R&D and Engineering Experience

Digiflow | Software Engineering Intern 07/2018-09/2018
Casablanca, Morocco (Remote)

- Developed enterprise workflows with AgilePoint, including expense-report processes and AI-based process routing.

Siprof S.A | Software Engineering Intern 06/2017-07/2017
Berrchid, Morocco

- Built VBA tools for production-defect monitoring and asset management; designed supporting databases and visualizations.

Skills and Languages

Programming: C++, Python, Java, VBA

Cryptography: SSE, structured encryption, encrypted multi-maps, ORAM / Path ORAM, TFHE, homomorphic encryption

Research: Clusion, cryptographic prototyping, benchmarking, leakage/performance analysis, probability, optimal stopping

Languages: Arabic, French, English